QR Code in KiCad

QR codes offer a simple and compact way to encode information directly onto a printed circuit board (PCB). They can store data such as configuration details, version numbers, or links to specific documentation.

As projects become more complex and documentation harder to manage, embedding a QR code on the PCB provides a convenient reference point, allowing users to access accurate and relevant information whenever they need it.

Solution: Place a QR code on the silkscreen layer and link it to a non-electrical symbol. This makes the QR code part of the PCB design, just like any other component, while keeping it synchronized with the schematic.

What You'll Learn

In this tutorial, we'll cover how to:

- Create a custom symbol to represent the QR code.
- Generate a QR code and save it as an image.
- Convert the image into a footprint for use in a PCB design.
- Link the symbol to the footprint.

1. Create a Custom Symbol

First, we'll create a symbol to represent the QR code in the schematic. This allows us to associate the QR code footprint with a schematic symbol and preserve the connection when updating the PCB.

Steps:

- Open the **Symbol Editor** (Tools → Symbol Editor).

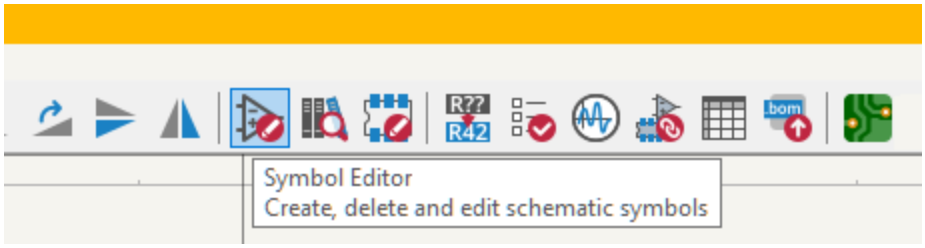


Figure 1: Symbol Editor

- Create a new library or add an existing one.
- Choose whether to create a **Global** or **Project** library.
- Name the symbol library, for example, `Graphics.kicad_sym`.
- Click **New Symbol**.

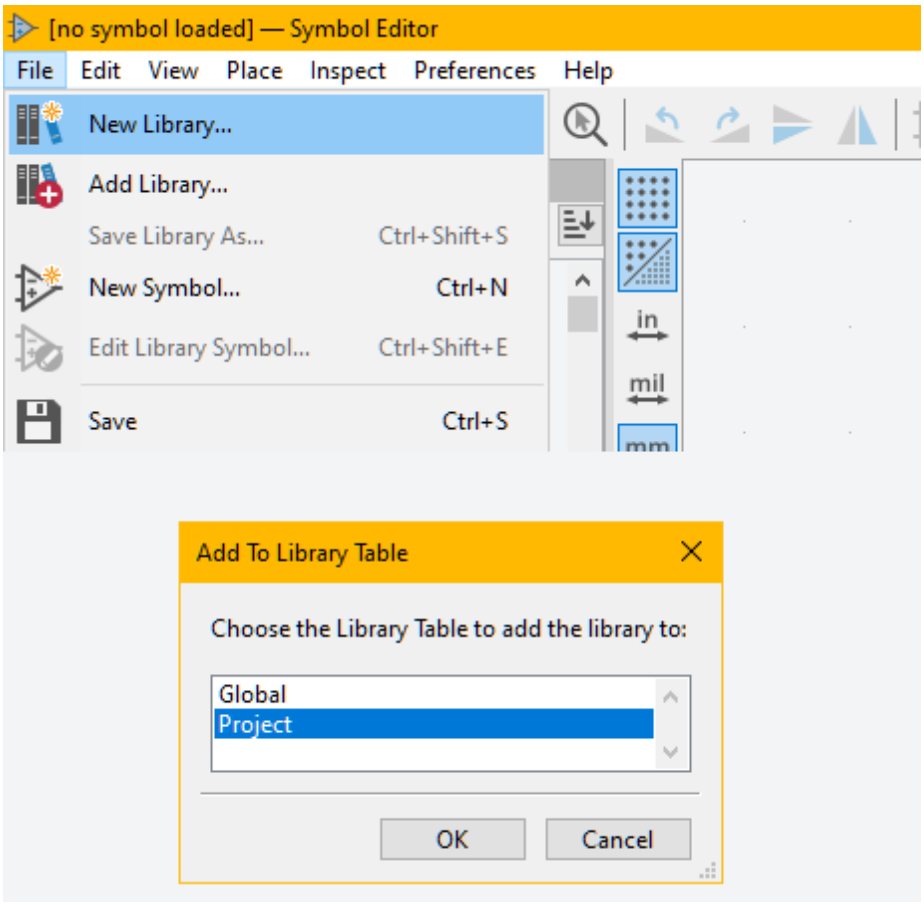


Figure 2: New Library

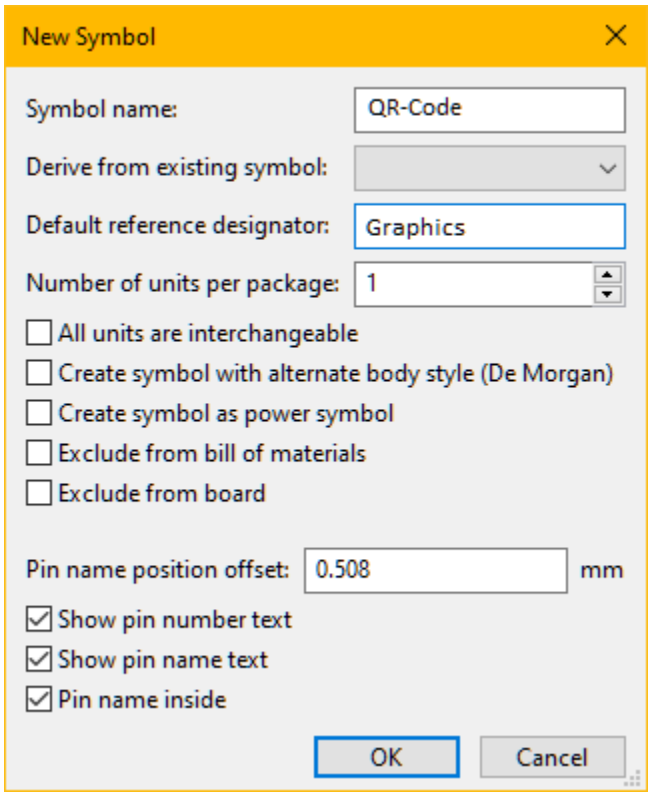


Figure 3: New Symbol

- Enter a name for the symbol, for example, `QR-Code`.
- Set the reference designator, for example, `Graphics`.
- Open **Symbol Properties** from the menu bar.

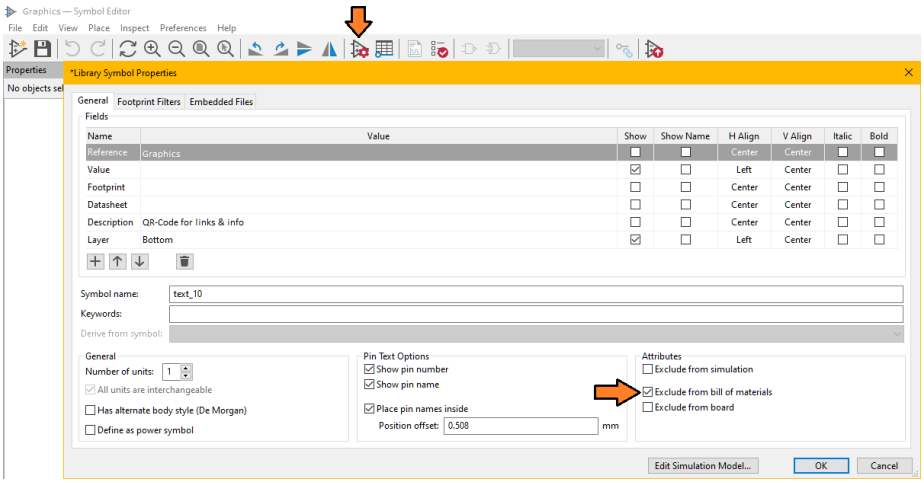


Figure 4: Symbol Properties

- Configure the symbol properties. The value is not important at this stage.
- Optionally, enable **Exclude from simulation**.
- Enable **Exclude from Bill of Materials** to avoid including this non-electrical item in the component list.
- Save the symbol, keeping the Symbol Editor open.
- Draw a simple symbol to represent the QR code.

This symbol acts purely as a graphical placeholder and does not affect the electrical functionality of the circuit.

2. Generate a QR Code

There are several websites that allow you to generate QR codes for free. Alternatively, you can easily create one using Python and the `qrcode` library.

Steps:

- Install the `qrcode` module:

  ```
  pip install qrcode
  ```

- Import the library in your Python script:

  ```
  import qrcode
  ```

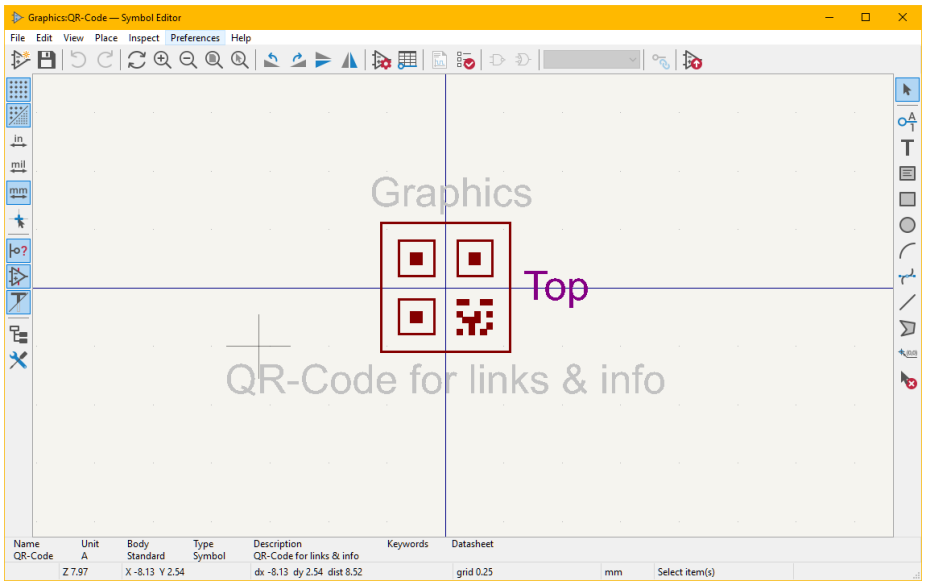


Figure 5: QR-Code Symbol

- Create a QR code object and define its parameters:

  ```
  qr = qrcode.QRCode(version=1, box_size=10, border=5)
  ```

- Add the information you want to encode:

  ```
  qr.add_data("YOUR INFORMATION")
  ```

- Fit the QR code to the required grid:

  ```
  qr.make(fit=True)
  ```

- Generate the QR code image:

  ```
  img = qr.make_image(fill_color="black", back_color="white")
  ```

- Save the image as a PNG file:

  ```
  img.save("YOUR FILE PATH/FILENAME.PNG")
  ```

You can also download our graphical interface (GUI) for generating QR codes from the following repository:

QR Code GUI – D+D Solve Labs

3. Create a Footprint from the QR Code Image

Once the QR code image has been generated, we can convert it into a KiCad footprint.



Figure 6: QR-Code GUI

Steps:

- Open the **Image Converter** from the KiCad project menu.
- Load the PNG image you have just created.
- Select the **Black & White Picture** tab.
- Set the desired **Output Size**.
- Select the footprint layer where the QR code should be placed, for example, `F.Silkscreen`.
- Click **Export to File...**.
- Open the generated footprint in the Footprint Editor.
- Select the autogenerated text item (usually `G`) and move it to a layer other than `F.Silkscreen`, for example, `F.Fab`. This prevents the generated label from appearing alongside the QR code on the silkscreen.

4. Link the Symbol to the Footprint

Now we need to associate the newly created footprint with the schematic symbol.

Steps:

- Return to the Symbol Editor.
- Open the **Symbol Properties** dialog.
- Assign the footprint you created, for example, ‘qrcode:qrcode

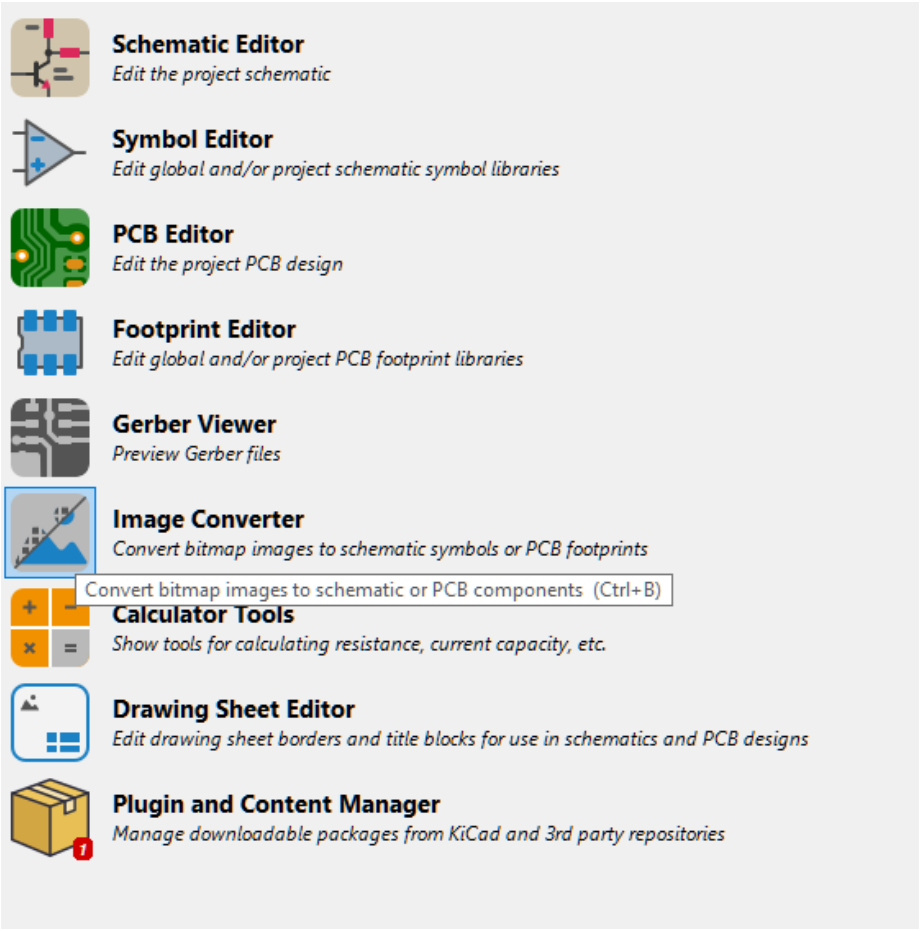


Figure 7: KiCad Image Converter

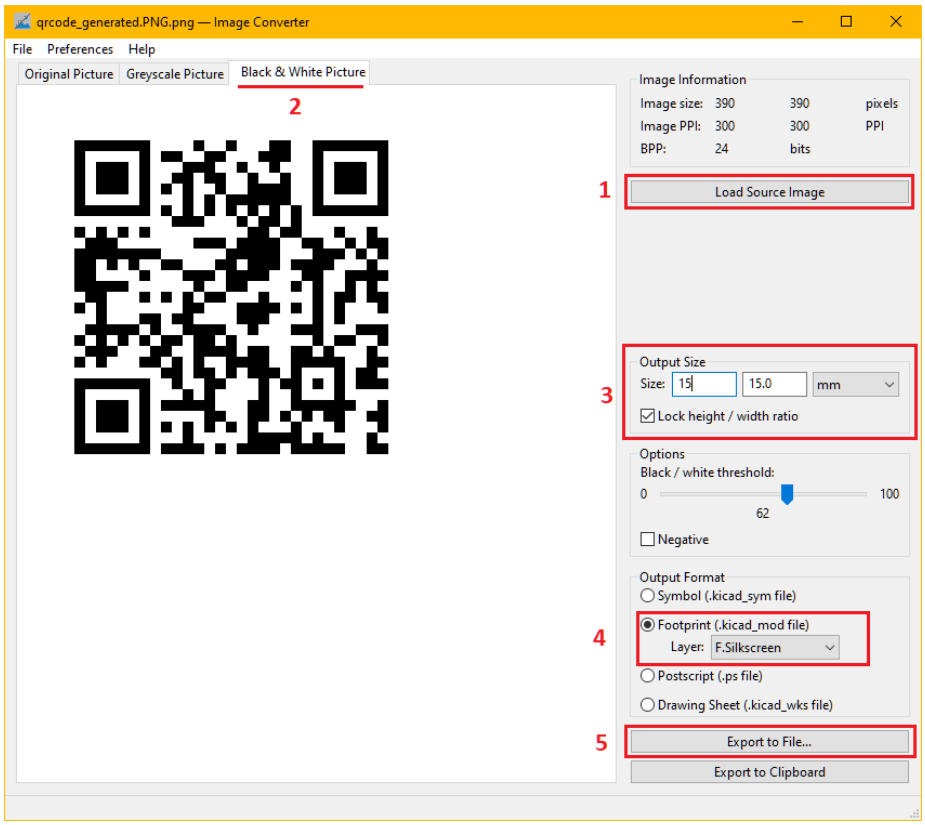


Figure 8: KiCad Image Converter Settings

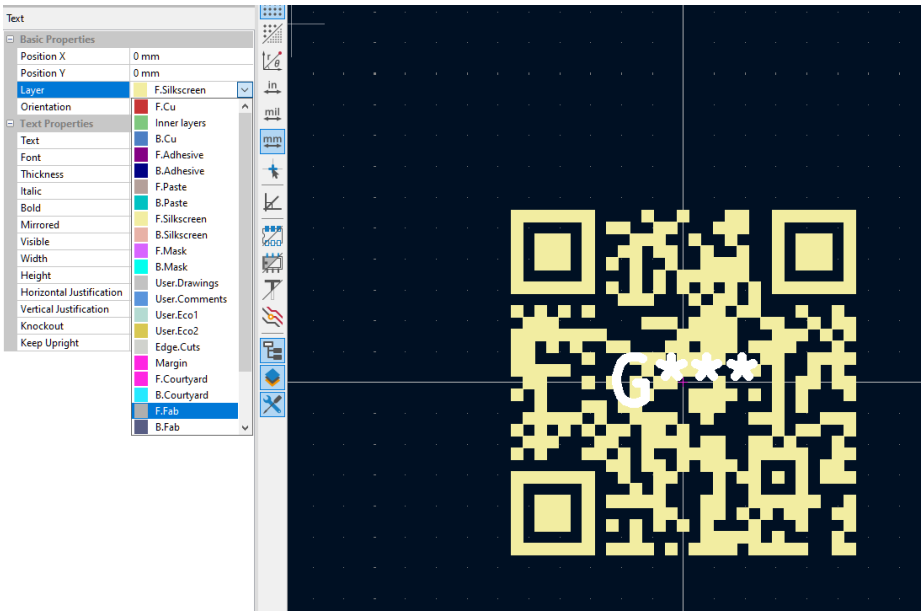


Figure 9: QR Code Footprint

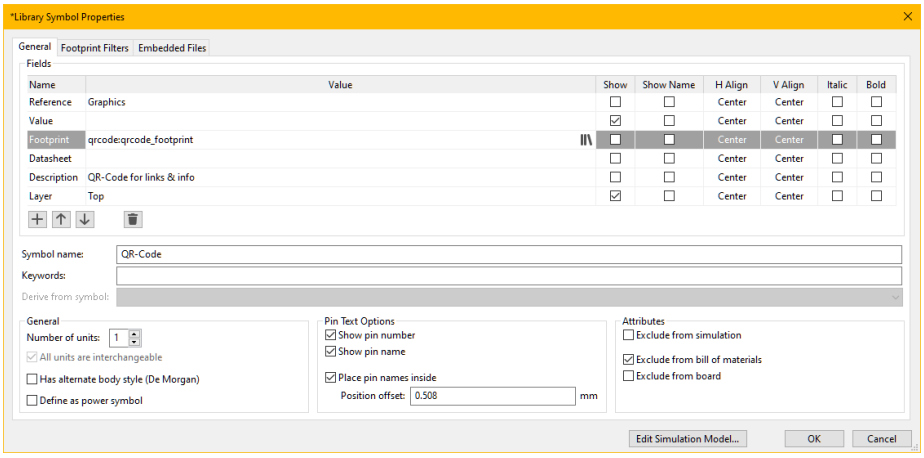


Figure 10: Symbol Properties